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Tensile properties of fumed silica filled polydimethylsiloxane networks

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ABSTRACT

Tensile properties of fumed silica filled hydroxylated polydimethylsiloxane (PDMS) networks were investigated in the current work. Similar to unfilled bimodal networks, unimodal networks filled with concentrated fumed silica exhibit non-Gaussian effect and improved ultimate tensile properties. The concept of "hierarchical network" was proposed to depict the networks exhibiting non-Gaussian effect at sufficiently high strains. It was found that the reinforcing effect originates from both the effective volume effect from filler volume and polymer–filler interaction and the synergistic effect between network chains within the "hierarchical network". Experimental results showed that filler's dispersion, concentration, specific surface area and surface chemistry have a great influence on the tensile properties, which could be interpreted by analyzing the variation of both effective volume effect and synergistic effect.
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1. Introduction

Polydimethylsiloxane (PDMS) rubber networks display a combination of very interesting properties such as extremely low glass transition temperature, high thermal and UV stability, low surface energy, excellent insulating properties, physiological inertness and biocompatibility [1,2]. They have potential applications in many fields such as elastomeric seals, adhesives, coatings, plastic surgery, etc. However, the poor mechanical properties significantly limit their application. In order to improve the mechanical properties, various fillers have been introduced, among which fumed silica is one of the most effective one. Moreover, PDMS network is a model network [3,4] to study reinforcing mechanisms, thanks to the controllable structure and the absence of self-reinforcing effect from, for example, strain-induced crystallization [5,6].

Networks are synthesized from single polymer chains by chemical crosslinking reaction. Therefore, the response of single polymer chain to external stress plays a fundamental role in the final mechanical properties of networks. Force spectroscopy for single polymer chains investigated by atomic force microscopy (AFM) showed that polymer chains exhibit Gaussian behavior from being taut to being straightened while exhibit non-Gaussian behavior when reaching their limited extensibility [7,8]. The former behavior was generally attributed to the conformational entropy loss caused by simple rotation of skeletal bonds and the later mainly to the enthalpic change due to the distortion of bond angles [9,10].

For unfilled networks, rearrangement and orientation of polymer chains proceed with increasing extension [11,12]. Part of network chains start to participate in transferring stress since they are stretched taut until disentanglement or rupture occurs (we name these chains as "transfer-chains"). Only Gaussian effect (or, linear behavior) is observed on the Mooney–Rivlin plots for unfilled unimodal networks (Fig. 1), suggesting that nearly all transfer-chains are in the Gaussian behavior regime before network breakup occurs thanks to the non-affine deformation [13]. It has been confirmed that the modulus of these networks decreases with chain length [14,15], increases with crosslinking functionality [15–17], and has nothing to do with dangling chains [18,19]. Compared with unfilled unimodal networks, unfilled bimodal networks show obvious non-Gaussian effect (non-linear upturn behavior) at sufficiently large strains and exhibit much improved ultimate tensile properties (refer to Fig. 1). The variance of modulus in the Gaussian regime can be attributed to the change of network structure and the non-Gaussian effect is unanimously attributed to the limited extensibility of short transfer-chains [13,20]. Mark and Tang [21] found that too many or too few short chains preclude the non-Gaussian effect and there exists an optimum mole ratio between short chains and long chains for bimodal networks to exhibit optimum ultimate tensile properties. Experiments [13,22,23] showed that "upturn strain" (i.e. the strain at the start of deviation from linearity on modulus plots, α_u) decreases with the mole ratio of short chains or with the increasing length difference between short chains and long chains. Research results [20,24] showed that trimodal networks can also exhibit non-Gaussian effect and improved ultimate tensile properties.

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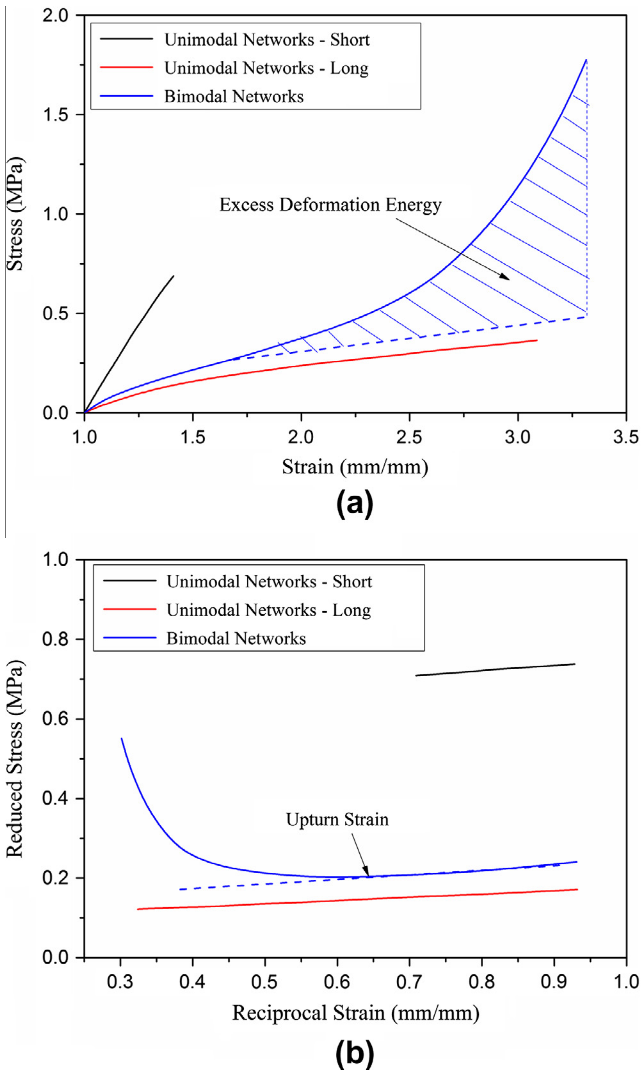

Fig. 1. Typical stress–strain curves (a) and Mooney–Rivlin plots (b) for unimodal and bimodal networks (referred to literature [23]). The shadow area corresponds to the excess deformation energy and dotted stress–strain curve belongs to the idealized unimodal networks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

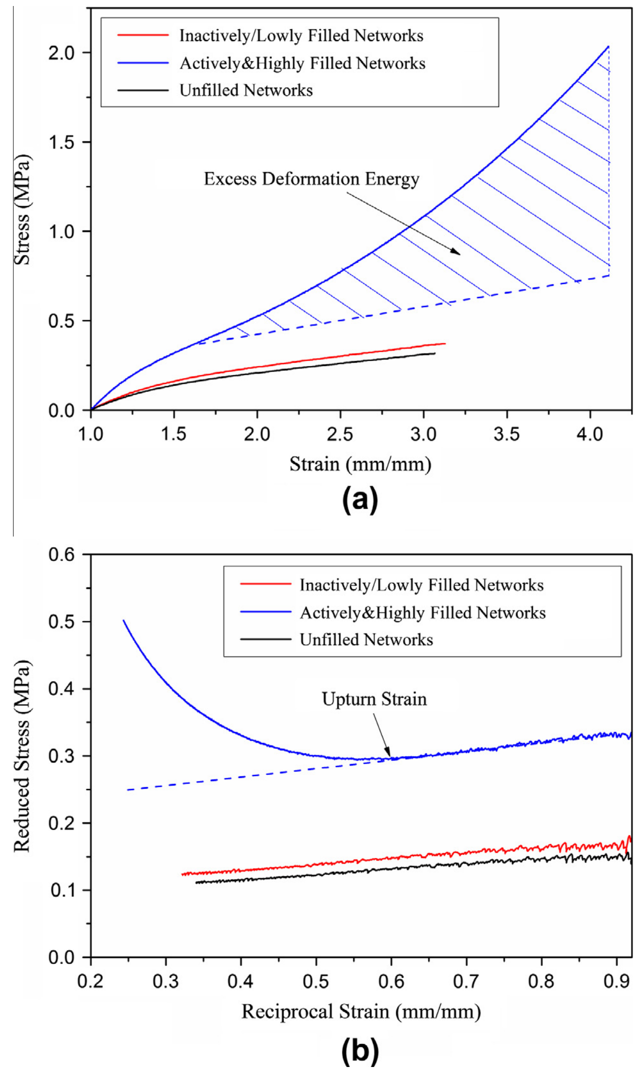

Fig. 2. Typical stress–strain curves (a) and Mooney–Rivlin plots and (b) for unfilled, inactively and actively (lowly and highly) filled networks (referred to literature [25,40]). The shadow area corresponds to excess deformation energy and dotted stress–strain curve belongs to the idealized unimodal networks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The behavior of network chains within filled (unimodal) networks is much more complicated. On top of the behaviors discussed above, the influence caused by filler particles is innegligible or even dominant. For networks filled with inactive fillers like glass spheres [25], Gaussian effect is normally observed (Fig. 2). The chain behavior is mainly affected by the excluded volume effect (or, hydrodynamic effect [26]) from filler particles. And the augmented modulus can be simply characterized by Smallwood–Einstein and other equations using filler's volume fraction [27–29]. For networks filled with active fillers like fumed silica with abundant silanol groups on the surface, chain behavior is affected not only by the excluded volume effect, but also by polymer–filler interaction. Thanks to the adsorption [30–32] and entanglement [33,34], polymer chains are divided into shorter segments. AFM results [35,36] showed that segments can participate in transferring stress and exhibit non-Gaussian behavior before desorption occurs. Thus, the tensile properties of networks filled with active fillers are determined not only by transfer-chains, but also by "transfer-segments". It is obvious that transfer-segments behave differently from transfer-chains under extension. On one hand, segments are more likely to desorb from the particle surface than to rupture at the backbone due to the large energy difference. Specifically, the desorption energy originated from hydrogen bond and van der Waals "bond" (ca. 10–40 kJ/mol [37]) is much smaller than the rupture energy of scission of Si–O bond (361 kJ/mol [2]). On the other hand, shorter segments would be more possible to participate in transferring stress and exhibit non-Gaussian behavior at a lower strain level, just as the short chains in bimodal networks. Similar to the bimodal networks having too few short chains, networks filled with diluted active fillers are also expected to exhibit Gaussian effect. Because the network chains are merely affected by the "effective volume effect" originated from filler volume and the change of network structure caused by polymer-filler interaction [38]. The modulus can be simply characterized by the equations for inactively filled networks using filler's effective volume fraction [39]. Highly and actively filled networks have been demonstrated to show marked non-Gaussian effect and much improved ultimate tensile properties as bimodal networks (see Fig. 2) [40]. Exploring the mechanisms of filler reinforcement has being one of the hot topics in the rubber fields for decades.

Until now, the reinforcing mechanisms of active fillers to rubber networks have not yet been completely clarified. Our present work is aimed at exploring the mechanisms and interpreting the effect of fumed silica, including dispersion, concentration, specific surface area and surface chemistry, on the tensile properties of filled PDMS networks.

2. Experimental

2.1. Materials

The α, ω-hydroxylated PDMS with weight average molecular weight of 80,300 g/mol and polydispersity index of 1.35 was obtained from WACKER (Germany). All kinds of fumed silica were provided by Evonik Degussa (Germany), which were labeled according to their trade names and part of their specifications are listed in Table 1. Specifically, R106 and R8200 were surface-modified from A300 using octamethylcyclotetrasiloxane (D_4) and hexamethyldisilazane (HMDS). Tetrafunctional crosslinker tetraethoxysilane (TEOS) and catalyst dibutyltin dilaurate (DBTDL) were used to synthesize PDMS networks. They were both supplied by TianJinBodi Corporation, China. The solvent of xylene (A.R., analytical reagent) were purchased from Beijing Chemical Works, China. All materials were used as received without further treatment.

2.2. Synthesis of networks

Fumed silica particles were mechanically blended with PDMS on an electronic controlled three-roll mill (EXAKT80E, Germany). Suspensions were aged in a vacuum oven at 100 °C for 24 h and then cooled down naturally. Unfilled and fumed silica filled PDMS networks were synthesized by the solution curing method as described below. A measured amount of suspensions was dissolved into xylene by stirring to form homogeneous solutions. The apparent viscosity of dissolved solutions is about 1 Pa s. Then a certain amount of TEOS and DBTDL were added into solutions and the vigorous stirring continued for another 5 min. After removal of bubbles in a vacuum oven, solutions were cast into Teflon molds. Finally, after curing at room temperature for 24 h, cured sheets were torn off the molds and put into an oven of 60 °C for one week. The obtained rubber sheets with the thickness of ca. 2 mm were used for tensile tests and other experiments.

2.3. Tensile tests

The specimens for tensile tests were stamped out from vulcanized rubber sheets using a dumbbell-shaped cutter (with a 4 mm width and 20 mm initial gauge length). The tests were performed at room temperature on a tensile testing machine CMT-4104 (SANS, China) at the crosshead rate of 50 mm/min. Loaded force was measured by a force sensor and elongation was recorded by a big deformation extensometer. At least five specimens were tested for each sample. Material parameters of ultimate tensile strength (f^*_m) and break elongation (α_m) can be directly obtained from stress–strain ($f^* - \alpha$) curves. The area under each stress–strain curve corresponds to the network breakup energy (E_b), which can be used as a measure of material toughness [23]. Meanwhile, elastic modulus or reduced stress can be obtained from the well-known Mooney–Rivlin equation: $[f^*] = 2C_1 + 2C_2\alpha^{-1}$, where constant $2C_1$ is taken as the limit modulus of a phantom network and $2C_2$ is regarded as a measure of the extent of non-affine deformation [14,41].

Table 1
Specific characteristics of different kinds of fumed silica.

AEROSIL silica	BET surface[a] (m^2/g)	Particle size[b] (nm)	Compact density[b] (g/l)	Modification agent[b]
A90	76	20	80	None
A200	170	12	50	None
A300	291	7	50	None
R106	256	7	50	D_4
R8200	161	7	140	HMDS

[a] Determined by the instrument in our lab.
[b] Supplied by the supplier Evonik Degussa.

2.4. Equilibrium swelling

Pieces of vulcanized rubber sheets (m_1) were immersed into toluene and kept within the solvent for 72 h at room temperature. After that, the swollen specimens were wiped gently with a filter paper to remove unabsorbed solvent and then their weights were measured as m_2. The amount of absorbed toluene was determined by the weight difference before and after swelling, $m_2 - m_1$. Equilibrium swelling ratio (q_{es}) was employed to simply reflect the degree of crosslinking (including both chemical crosslinking and physical adsorption), which was defined as the ratio of the mass of absorbed solvent to the mass of pure rubber: $q_{es} = \frac{m_2 - m_1}{m_1(1-\phi_m)}$, where ϕ_m is the mass fraction of fumed silica. Considering the simple procedure to obtain q_{es}, one specimen was tested for each sample.

2.5. Characterization

The surface chemistry of fumed silica was examined by infrared spectroscopy (IR) using a Spectrum One IR spectrometer (PerkinElmer, USA). The scanning range was between 4000 and 500 cm^{-1} with a resolution of 4 cm^{-1}.

The dispersion of fumed silica in the PDMS networks was studied by inspecting the fracture surface of the specimens on a scanning electron microscope (SEM, HITAHI S-4800, Japan) apparatus. The surfaces were coated by gold prior to observation. The acceleration voltage was chosen as 6 kV.

3. Results and discussion

3.1. Mechanisms of reinforcement

According to the classic elastic theory [42], the value of modulus is proportional to the number density of transfer-chains per unit initial cross-sectional area. The modulus of unfilled unimodal networks is quite small, suggesting that only a small part of network chains participate in transferring stress due to the topology of three-dimensional network structure. It is obvious that the network breakup energy is composed of almost merely entropic energy from transfer-chains.

For modulus of unfilled multimodal networks, the linear behavior at low strains still conforms with the classical elastic theory, whereas the upturn behavior at high strains suggesting the existence of a new type of network structure. Based on the conclusions mentioned in the introduction section, we propose using "hierarchical network" concept to depict the networks that can exhibit non-Gaussian effect at sufficiently high strains, where bimodal network is the simplest case. Two essential conditions have to be met for the "hierarchical network" structure to form: firstly, network chains must have multimodal length distribution with large enough length difference; secondly, the concentrations of both shorter chains and longer chains must be above certain percolation thresholds. Only if these two conditions are met, can the networks overcome non-affine deformation to exhibit non-Gaussian effect.

In the "hierarchical network", shorter chains can reach their limited extensibility at a lower stain level and then be ruptured to act as rupture nuclei, while longer chains inhibit the growth of rupture nuclei to retard the occurrence of breakup. This mechanism is also known as synergistic effect [43], which is mainly determined by the length difference and component of network chains. The synergistic effect can be characterized by the "excess deformation energy" compared with the idealized unimodal networks having the same modulus at low strains, as the shadow area denoted in Fig. 1(a). The network breakup energy consists of not only entropic energy, but also enthalpic energy from transfer-chains.

Analogue to multimodal networks, unimodal networks filled with concentrated active fillers also show non-Gaussian effect at sufficiently large strains. The augmented modulus in the linear regime can still be attributed to the filler's effective volume effect. And the upturn of modulus at large strains indicates the existence of "hierarchical network" structure (refer to Fig. 3), which takes formation only when the concentration of filler particles exceeds a percolation concentration, or directly, when the component of segments exceeds a percolation threshold. The "hierarchical network" sets a stage for the segments to exhibit non-Gaussian behavior before desorption occurs and for the networks to show non-Gaussian effect. The resulting synergistic effect can also be characterized by the excess deformation energy compared with the idealized unimodal networks as denoted in Fig. 2(a). The network breakup energy consists of both entropic energy and enthalpic energy from both transfer-segments and transfer-chains. In short, the reinforcing effect of these actively filled networks mainly originates from filler's effective volume effect and synergistic effect. Other factors having influence on the network structure, for example, transformation of desorbed segments into transfer-chains, extent of filler agglomeration [44,45], occluded rubber within huge agglomerates [46,47], etc., can all be regarded as the factors that affect the effective volume effect and synergistic effect.

3.2. Effect of dispersion on tensile properties

Filler's dispersion has a great influence on the final mechanical properties of filled rubber networks [48–50]. To study the effect of dispersion states on tensile properties, PDMS networks were synthesized from differently dispersed suspensions with10 phr of R106 silica. Samples were labeled as "45 μm", "15 μm" and "15 N" according to the minimum gap or maximum stress between rolls. It has been demonstrated [51] that the external force has a huge influence on the structure of agglomerations. Obviously, sample "15 N" has the best dispersion state and "45 μm" has the worst.

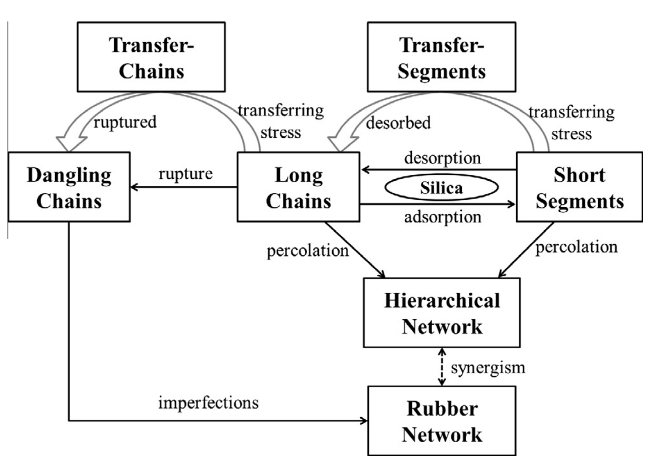

Fig. 3. Establishment of hierarchical network structure within highly and actively filled networks.

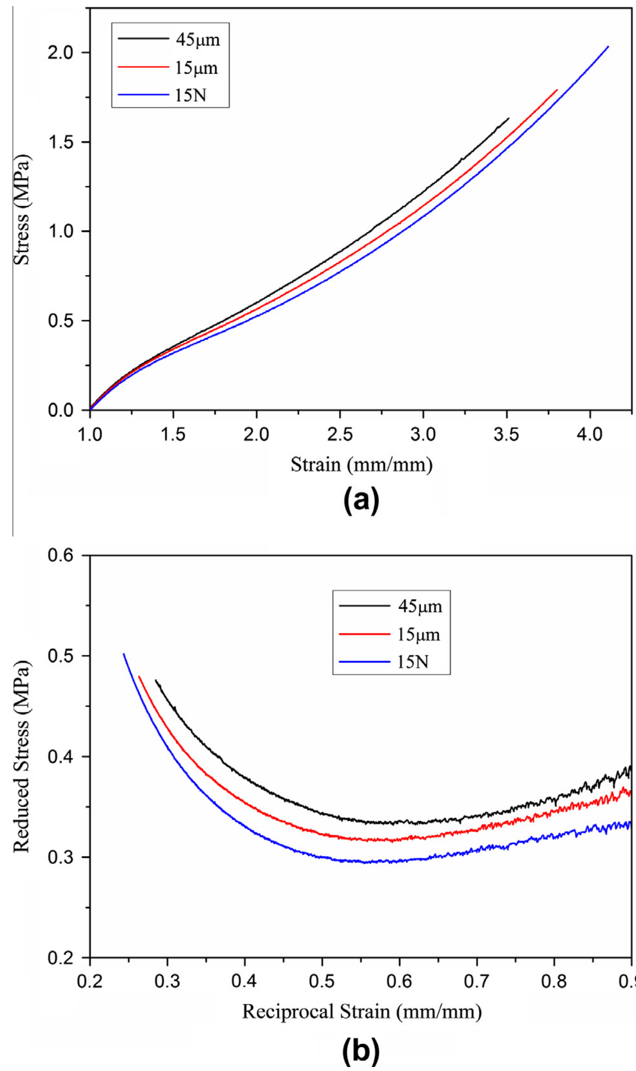

Fig. 4. Stress–strain curves (a) and Mooney–Rivlin plots and (b) for networks of different dispersion states. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Simple tension results for networks of different filler dispersion states are shown in Fig. 4 and listed in Table 2. Marked non-Gaussian effect was observed for all samples, suggesting the existence of "hierarchical network" structure within the filled networks. From Mooney–Rivlin plots we see that modulus decreases to various extents with improved dispersion. The decrease of modulus can be attributed to the reduction of transfer-segments caused by the change of network structure. Specifically, some segments between filler particles become longer when the extent of filler agglomeration alleviated. The lengthened segments would participate in transferring stress and reach their limited extensibility at a larger strain. This is somewhat similar to the replacement of short chains by longer short chains in the bimodal networks. The existence of lengthened segments can be confirmed from the larger upturn strains and equilibrium swelling ratio as listed in Table 2. However, from stress–strain curves we see that ultimate tensile strength, break elongation and breakup energy (also excess deformation energy) increase with improved filler dispersion. This can be interpreted that the more homogeneous distribution of segments within the "hierarchical network" contributes to the delayed

Table 2
Tensile properties and equilibrium swelling ratios for unfilled and filled networks.

Samples	f_m (MPa)	α_m (%)	E_b (J/mm^3)	α_u	q_{es}
45 μm (R106, 10 phr)	1.63 ± 0.07	256 ± 9	1.97 ± 0.08	1.41 ± 0.05	2.46
15 μm (R106, 10 phr)	1.83 ± 0.05	286 ± 6	2.33 ± 0.06	1.54 ± 0.02	2.51
15 N (R106, 10 phr)	2.04 ± 0.04	308 ± 5	2.77 ± 0.05	1.59 ± 0.04	2.57
PDMS	0.36 ± 0.01	207 ± 6	0.49 ± 0.02	–	3.38
1 phr (R106)	0.43 ± 0.01	213 ± 3	0.61 ± 0.02	–	3.05
3 phr (R106)	0.70 ± 0.03	227 ± 10	0.83 ± 0.04	1.96 ± 0.02	2.92
5 phr (R106)	0.92 ± 0.02	239 ± 6	1.17 ± 0.03	1.75 ± 0.03	2.80
10 phr (R106)	2.04 ± 0.04	308 ± 5	2.77 ± 0.05	1.59 ± 0.04	2.57
A90 (10 phr)	1.33 ± 0.05	279 ± 10	1.79 ± 0.06	1.64 ± 0.02	2.79
A200 (10 phr)	1.84 ± 0.09	348 ± 14	2.68 ± 0.10	1.54 ± 0.03	2.73
A300 (10 phr)	2.19 ± 0.12	368 ± 11	3.22 ± 0.14	1.49 ± 0.06	2.70
R106 (10 phr)	2.04 ± 0.04	308 ± 5	2.77 ± 0.05	1.59 ± 0.04	2.57
R8200 (10 phr)	1.51 ± 0.11	307 ± 16	2.15 ± 0.12	1.82 ± 0.05	2.85

occurrence of network breakup. This renders a larger synergistic effect and thus better ultimate tensile properties.

3.3. Effect of concentration on tensile properties

The effect of concentration on the tensile properties of filled networks has been most extensively investigated [52–54]. To investigate this effect, a series of networks filled with increasing concentration of R106 were prepared and labeled with the concentrations.

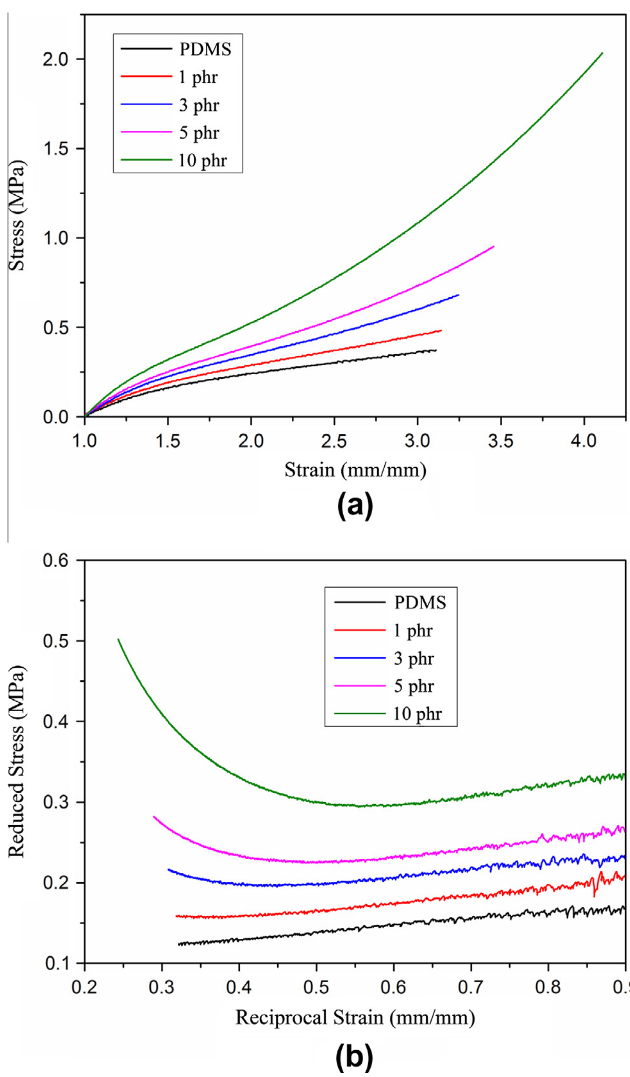

Fig. 5. Stress–strain curves (a) and Mooney–Rivlin plots and (b) for networks filled with variant concentrations of R106. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

It can be seen from Fig. 5 and Table 2 that all mechanical parameters increase with filler concentration. At low concentrations (such as 1 phr), modulus exhibits almost linear behavior similar to unfilled networks and tensile properties show very limited enhancement. This is because the "hierarchical network" structure fails to form and filler's effective volume effect makes the major contribution. At higher concentrations (above 3 phr), modulus curves begin to show obvious non-Gaussian effect starting at lower strains and tensile properties become more apparently enhanced. This experimentally confirms that the "hierarchical network" forms only when the filler concentration exceeds a percolation threshold. It can be expected that this filler percolation concentration is correlated with the value (e.g. 2–3 phr) obtained from the rheological tests as presented in our previous work. The reinforcing effect originates from both effective volume effect and synergistic effect for highly filled networks. Specifically, the effective volume effect includes the excluded volume effect from filler particles and the change of network structure induced by polymer–filler interaction. Meanwhile, as filler concentration increases, the number density of segments increases and the segmental length shortens due to the decreased particle distance and aggravated filler agglomeration (see Fig. 6 (a and b)), which provides a stronger "hierarchical network" exhibiting larger synergistic effect. The decrease of upturn strain and equilibrium swelling ratio (see Table 2) confirms the change of segments in the networks. At very high concentrations (above ca. 30–40 phr), from similar experimental results in Fig. 7 we see that the increase of ultimate tensile strength slows down and the break elongation decreases, indicating the networks starts to transform from rubbery elastomers to rigid ormosils [55].

3.4. Effect of specific surface area on tensile properties

It is well-known that filler's specific surface area has a huge influence on the mechanical properties of filled networks [40,52,53]. In this section, networks filled with pristine fumed silica of different specific surface areas at a concentration of 10 phr were prepared and tested. Samples were labeled with the trade names of fumed silica: A90, A200 and A300.

It can be seen from Fig. 8 and Table 2 that all mechanical parameters increase with specific surface area. This reinforcement should have almost the same explanations as that in the concentration section. Thanks to the constant silanol group density [56,57] and larger surface area, the number of silanol group increases with specific surface area (as shown in Fig. 9 that the peak corresponding to Si–OH group around 3500 cm^{-1} increases) and the number of